

PES UNIVERSITY

Department of Computer Science and Engineering

SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT

DELIVERABLE 2 – REQUIREMENTS

Railway Reservation System – Train Scheduling, Seat Allocation and Ticket Management

Item	Details
Project Title	Railway Reservation System – Train Scheduling, Seat Allocation and Ticket Management
Group Number	Team 14
Customer / Stakeholder	Railway Reservation / Ticketing Operations
Team Members	Bhadra K S (PES1UG24CS111) / Kshitij Prabhu (PES1UG24CS914) / Catherine D (PES1UG24CS120) / K Pranav (PES1UG24CS819)
Course	UE24CS341A – Software Engineering
Semester	V Semester
Development Methodology	Agile
Version	1.0

2. REQUIREMENTS

This package covers the Software Requirements Specification (SRS), the initial Validation Specification and the Project Plan for the Railway Reservation System, building on the approved Deliverable 1 synopsis.

2.1 Introduction

Purpose: This SRS defines the functional, non-functional and security requirements of the Railway Reservation System, which digitizes train search, seat availability, reservation booking, PNR generation, cancellation and waitlist handling, and administrative management of train and schedule data.

Audience: Development team members (Bhadra K S, Kshitij Prabhu, Catherine D, K Pranav), project mentors/evaluators, and the customer/stakeholder representing Railway Reservation / Ticketing Operations.

Scope of this document: This SRS covers the requirements baseline for Sprint 1 onward as defined in the approved synopsis short-term plan. It does not cover physical train operations, signalling, or payment-gateway settlement, which remain outside the initial system scope.

References: Deliverable 1 – Synopsis/Project Proposal (Railway Reservation System, Team 14); Software Engineering – Level 3 Student Submission Guide, PES University.

2.2 Overall Description

Product perspective: The system is a new, standalone reservation application developed in C and C++, replacing a fragmented/manual reservation process with a structured, modular solution.

Product functions (summary): passenger registration/login and profile management; train search and schedule/route/fare display; seat availability checking and allocation; reservation booking with PNR generation; cancellation, refund calculation and waitlist handling; administrator management of trains, routes, schedules, coaches/seats and reservation records; operational reporting for railway management.

User characteristics: Passengers are non-technical, occasional-to-frequent users interacting through a menu-driven interface. Reservation staff and administrators are trained users who interact more frequently and require higher-privilege operations.

Operating environment: Developed and run on a standard Linux/Windows environment using GCC/G++, with persistent storage via local files or an embedded database, version-controlled with Git/GitHub.

Constraints: The system must be implemented in C and C++ using modular/object-oriented design; must use Git/GitHub for version control with individually attributable commits; must use Agile sprints as defined in the project plan.

Assumptions and dependencies: Train, route and initial schedule data are seeded/maintained by an administrator; payment-gateway settlement is out of scope, so booking confirmation depends on fare calculation and passenger-detail validation rather than real payment processing.

2.3 Scope

In Scope	Out of Scope
Passenger registration, login and profile management	Physical railway operations and real-time signalling
Train search by source, destination and journey date	Train movement control
Train schedule, route and fare display	Payment-gateway settlement
Seat availability checking and seat allocation	Integration with live railway infrastructure
Reservation booking and PNR generation	

In Scope	Out of Scope
Cancellation, refund calculation and waitlist/status handling	
Administrator management of trains, routes, schedules, coaches/seats and reservation records	
Operational reporting (booking statistics, seat occupancy)	

2.4 User Classes

User Class	Privileges	Characteristics
Passenger / Customer	Register/login, search trains, view schedule/fare/availability, book, view PNR/status, cancel own tickets	Non-technical, occasional-to-frequent use
Reservation Staff	Create/update bookings on behalf of passengers, process cancellations, manage waitlist and reservation records	Trained user, frequent operational interaction
System Administrator	Add/update/delete train, route, schedule, coach and seat data; manage user and reservation records	Higher-privilege, trained user
Railway Operations / Management	View booking statistics, seat occupancy, schedules and reservation reports	Trained user, periodic/reporting use

2.5 External Interfaces

User Interface: A role-based, menu-driven console interface for Passenger, Reservation Staff and Administrator roles, consistent with a C/C++ implementation.

Software Interfaces: Persistent storage of train, schedule, passenger and reservation records via local data files or an embedded database accessed through the C/C++ data-access layer.

Communication Interfaces: The initial system is a standalone application with no external network interface; integration with a live payment gateway or railway infrastructure is explicitly out of scope.

Hardware Interfaces: Standard desktop/server hardware capable of running the compiled C/C++ executable; no specialised hardware is required.

2.6 Analysis Models

Use case diagram: Actors — Passenger/Customer, Reservation Staff, System Administrator, Railway Operations/Management — connected to the use cases listed below. Students should insert a clean use-case diagram (e.g., drawn in Draw.io/StarUML) showing these actor-to-use-case relationships.

Use Case ID	Use Case	Primary Actor	Precondition	Outcome
UC-01	Register / Login	Passenger	Application available	Authenticated session
UC-02	Search Trains	Passenger	None	Matching trains displayed
UC-03	View Schedule, Route & Fare	Passenger	Train selected	Schedule/route/fare displayed
UC-04	Book Ticket	Passenger	Logged in; valid passenger details	Reservation created; PNR generated
UC-05	Check PNR / Ticket Status	Passenger	PNR exists	Current status displayed

Use Case ID	Use Case	Primary Actor	Precondition	Outcome
UC-06	Cancel Ticket	Passenger	Valid, active booking exists	Ticket cancelled; refund calculated
UC-07	Process Waitlist	System / Reservation Staff	Cancellation or new capacity occurs	Waitlist re-ordered/updated
UC-08	Manage Bookings	Reservation Staff	Staff authenticated	Booking created/updated on passenger's behalf
UC-09	Manage Trains, Routes & Schedules	System Administrator	Admin authenticated	Train/route/schedule data updated
UC-10	Manage Reservation Records	System Administrator	Admin authenticated	Reservation records updated/corrected
UC-11	View Operational Reports	Railway Operations / Management	Authenticated	Report displayed

2.7 System Features / Functional Requirements

Each requirement below is uniquely identified, concise, verifiable and mapped to one of the four owned functionality areas from the synopsis: registration/profile (Bhadra K S), train search/schedule (Kshitij Prabhu), seat/booking/fare/PNR (Catherine D), and cancellation/waitlist/administration (K Pranav).

2.7.1 Feature: Passenger Registration, Login & Profile Management

ID	Requirement	Priority	Verification
FR-001	The system shall allow a new passenger to register an account using a unique mobile number or email ID.	High	Functional test
FR-002	The system shall allow a registered passenger to log in using valid credentials.	High	Functional test
FR-003	The system shall allow an authenticated passenger to create and update their profile (name, age, gender, contact details, ID-proof type and number).	High	Functional test
FR-004	The system shall reject registration attempts that use a mobile number/email already associated with an existing account.	High	Negative test

2.7.2 Feature: Train Search, Schedule & Route Management

ID	Requirement	Priority	Verification
FR-005	The system shall allow a passenger to search for trains by source station, destination station and journey date.	High	Functional test
FR-006	The system shall display all matching trains with train number, name, departure time, arrival time and duration.	High	Functional test

ID	Requirement	Priority	Verification
FR-007	The system shall allow a passenger to view the full schedule and route (ordered list of stopping stations) for a selected train.	Medium	Functional test
FR-008	The system shall display fare information for each available class for the selected train and journey.	High	Functional test

2.7.3 Feature: Seat Availability, Booking, Fare & PNR Generation

ID	Requirement	Priority	Verification
FR-009	The system shall display real-time seat availability for a selected train, date and class.	High	Functional test
FR-010	The system shall allow a passenger to enter passenger details (name, age, gender, ID proof) for each traveller in a booking.	High	Functional test
FR-011	The system shall calculate the total fare from the base fare, class and number of passengers before booking confirmation.	High	Unit + system test
FR-012	The system shall confirm a reservation only after all mandatory passenger details are valid and the calculated fare is accepted (payment-gateway settlement is out of scope).	High	Negative + positive test
FR-013	The system shall allocate a specific seat/berth number to each confirmed passenger at the time of booking confirmation.	High	Integration test
FR-014	The system shall generate a unique PNR number for every successfully created reservation.	High	Integration test
FR-015	The system shall maintain a booking history accessible to the passenger, showing past and upcoming reservations.	Medium	Functional test

2.7.4 Feature: Cancellation, Waitlist & Administrator Operations

ID	Requirement	Priority	Verification
FR-016	The system shall allow a passenger to cancel a booked ticket before the journey date.	High	Functional test
FR-017	The system shall calculate the applicable refund amount based on the defined cancellation policy and time of cancellation.	High	Unit test
FR-018	The system shall place a passenger on a waitlist, with a visible waitlist position, when no confirmed seats are available.	High	Functional test
FR-019	The system shall automatically promote the next waitlisted ticket to confirmed status when a confirmed seat becomes available due to cancellation.	High	Integration test
FR-020	The system shall allow a passenger to check the current status of a ticket (Confirmed, RAC, Waitlisted, Cancelled) using the PNR number.	High	Functional test
FR-021	The system shall allow an administrator to add, update or remove train, route and schedule information.	High	Functional test

ID	Requirement	Priority	Verification
FR-022	The system shall allow an administrator to manage coach and seat/berth configuration for each train.	Medium	Functional test
FR-023	The system shall allow reservation staff to create or update a booking on behalf of a passenger.	Medium	Functional test
FR-024	The system shall allow an administrator or reservation staff to manually correct/cancel a reservation record when required.	Medium	Functional test
FR-025	The system shall allow an administrator or railway-operations user to generate reports on booking statistics, seat occupancy and train-wise reservation counts.	Medium	Functional test

2.8 Non-Functional Requirements

ID	Category	Measurable Requirement	Verification
NFR-001	Performance	95% of train search queries shall return results within 2 seconds under the defined test load.	Performance test
NFR-002	Availability	The application shall be available at least 99% during the defined operating window, excluding planned maintenance.	Operational review
NFR-003	Usability	A new passenger shall be able to complete a basic booking in no more than 6 primary menu steps.	Usability walkthrough
NFR-004	Reliability	A reservation shall not be created twice if a passenger retries after a booking-confirmation timeout.	Negative/integration test
NFR-005	Maintainability	Business logic (search, booking, cancellation, waitlist) shall be separated into identifiable modules with automated unit tests.	Code review
NFR-006	Testability	Core services (seat allocation, fare calculation, PNR generation) shall expose testable interfaces independent of file/database I/O.	Unit/integration test
NFR-007	Portability	The core C/C++ modules shall compile and run without modification using the project's specified GCC/G++ toolchain.	Build verification

2.9 Security Requirements

ID	Security Requirement	Validation Approach
SEC-001	Passwords/credentials shall not be stored in plaintext.	Code/configuration review
SEC-002	Role-based access shall prevent passengers from accessing reservation-staff or administrator functions.	Authorization tests
SEC-003	The system shall validate and sanitize all user-supplied input (passenger details, search parameters) before processing.	Negative/security tests
SEC-004	The system shall reject authenticated operations when the session/credentials are invalid or expired.	Security test
SEC-005	Sensitive passenger identification data shall not be retained beyond what is necessary for reservation processing.	Design review

ID	Security Requirement	Validation Approach
SEC-006	Security-relevant events (failed logins, unauthorized attempts to access admin functions) shall be logged.	System test

2.10 Business Rules / Domain Rules

- Only trains with available confirmed seats or waitlist capacity can accept new bookings.
- A passenger cannot modify passenger details after a ticket is confirmed and a PNR is generated; only cancellation is permitted.
- Only reservation staff or administrators can manually correct or override a reservation record.
- Only administrators can add, update or remove train, route and schedule data.
- A booking must include at least one valid passenger entry.
- Cancellation and refund amounts shall follow the defined refund policy based on time-to-departure.
- Waitlisted tickets are confirmed strictly in order of waitlist position when a confirmed seat becomes available.

2.11 Glossary (Appendix A)

Term	Definition
PNR	Passenger Name Record — the unique reference number generated for each reservation.
RAC	Reservation Against Cancellation — a partially confirmed status pending full seat allocation.
Waitlist	A queue of passengers awaiting a confirmed seat, ordered by booking time/position.
Coach	A physical unit of a train containing a defined class and set of seats/berths.
Berth	An individual seat/sleeping position within a coach, uniquely numbered.
Fare	The calculated price of a reservation based on class, distance and number of passengers.

2.12 Field Layouts (Appendix B)

Field	Length	Data Type	Description	Mandatory
passenger_id	36	UUID/String	Unique passenger identifier	Y
name	100	String	Passenger full name	Y
age	3	Integer	Passenger age	Y
contact_number	15	String	Mobile number, used for login/registration	Y
id_proof_number	30	String	Identity proof reference number	Y
train_id	36	UUID/String	Unique train identifier	Y
train_number	10	String	Publicly displayed train number	Y
source_station	50	String	Journey origin station	Y
destination_station	50	String	Journey destination station	Y
schedule_id	36	UUID/String	Unique schedule/journey-date identifier	Y
coach_class	20	Enum/String	Coach class (e.g., Sleeper, AC 3-Tier)	Y

Field	Length	Data Type	Description	Mandatory
seat_number	10	String	Allocated seat/berth number	N (until confirmed)
pnr_number	10	String	Unique reservation reference	Y
status	20	Enum/String	Confirmed / RAC / Waitlisted / Cancelled	Y
fare_amount	10,2	Decimal	Calculated total fare	Y

2.13 Requirement Traceability Matrix — Initial (Appendix C)

At the requirements stage, architecture, design, code and test-case references are not yet available; they will be completed in Deliverables 3 and 4. The structure is established now so it can be filled in progressively.

Req ID	Requirement (brief)	Architecture Ref	Design Ref	Code/File Ref	Test Case ID	System Test ID
FR-002	Passenger login	TBD	TBD	TBD	TBD	TBD
FR-009	Seat availability check	TBD	TBD	TBD	TBD	TBD
FR-012	Booking confirmation	TBD	TBD	TBD	TBD	TBD
FR-014	PNR generation	TBD	TBD	TBD	TBD	TBD
FR-017	Refund calculation	TBD	TBD	TBD	TBD	TBD
FR-019	Waitlist promotion	TBD	TBD	TBD	TBD	TBD
SEC-002	Role-based access	TBD	TBD	TBD	TBD	TBD

2.14 Validation Specification – Initial Planning

At this stage, the validation specification is planning-focused: it explains how functional, security and non-functional requirements will eventually be validated, including planned penetration/security testing. This is not the final test report, which is produced in Deliverable 5.

Requirement Category	Planned Validation	Example
Functional	Black-box functional tests covering each system feature	Valid vs. invalid booking attempt
Security	Authentication/authorization and basic penetration/security testing	Passenger attempting to access an administrator endpoint/menu
Performance	Load/response-time testing	Concurrent train-search and booking requests
Usability	Task-based walkthrough	New passenger completes a booking end-to-end
Reliability	Failure and retry scenarios	Booking-confirmation timeout and retry
Unit	Automated unit tests with coverage reporting	Fare calculation, seat allocation, PNR generation logic

2.15 Project Plan

The project plan identifies and justifies the Agile lifecycle, tools, work breakdown, effort estimates and schedule, building on the short-term plan approved in the synopsis.

2.15.1 Work Breakdown Structure

Work Package	Tasks	Owner	Estimated Effort (person-days)	Output
WP1	Requirements, use cases, initial validation planning	All	5	SRS + initial Validation Spec
WP2	Architecture + UML design	Kshitij Prabhu / Catherine D	4	Architecture/Design
WP3	Registration/login + passenger profile module	Bhadra K S	6	Working feature
WP4	Train search + schedule/route management module	Kshitij Prabhu	6	Working feature
WP5	Seat availability, booking, fare & PNR module	Catherine D	7	Working feature
WP6	Cancellation, waitlist & administrator module	K Pranav	7	Working feature
WP7	Testing + CI/CD + security validation	All	6	Test/validation evidence
WP8	Final validation + consolidated report	All	4	Final report

2.15.2 Schedule (aligned to Sprints 1–6 from the synopsis)

Sprint	Planned Output	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, user registration/login	User stories + Git repository + working login
Sprint 2	Train search, route and schedule management	Working search demo + unit tests
Sprint 3	Seat availability, allocation and fare calculation	Demo + unit/integration test evidence
Sprint 4	Reservation workflow, passenger details and PNR generation	End-to-end booking demo + PR evidence
Sprint 5	Cancellation, waitlist/status and administrator functions	Functional validation + coverage/security evidence
Sprint 6	System integration, validation, defect fixing and release	Final test report + working demonstration

2.15.3 Tool Stack

Activity	Tool	Evidence to Submit
Programming	C and C++	Source code
Development	Visual Studio Code / GCC or G++	Build configuration
Version control	Git + GitHub	Repository URL, commit history
Agile planning	Jira / GitHub Projects	Backlog, user stories, sprint board
CI/CD	Jenkins	Pipeline screenshot/log
Containerization	Docker	Dockerfile + build/run evidence
Testing	GoogleTest (gtest) or equivalent	Test results

Activity	Tool	Evidence to Submit
Static analysis	SonarQube or similar	Analysis report
Design	Draw.io / StarUML / PlantUML	UML diagrams